Matthew Evans

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decentralized data management • open science & software
materials discovery • ab initio calculations

RESEARCH INTEREST

My background in computational materials science has left me with an overarching interest in the application of artificial intelligence, open source software, and data management infrastructure to accelerate and enhance scientific workflows for discovery in the chemical and materials sciences.

EDUCATION

2016–2023 **PhD Physics**, Theory of Condensed Matter Group, University of Cambridge

2015–2016 **MPhil Scientific Computing**, University of Cambridge, *Pass with distinction*

2011–2015 **MPhys Physics with Theoretical Physics**, University of Manchester, *First Class (Hons)*

SELECTED EXPERIENCE

2025– **Leverhulme Trust Early Career Research Fellow**

University of Cambridge, with Prof Dame Clare Grey FRS

- Co-creator and architect of *datalab*, open source data management software for sample tracking and characterisation, lab management, and machine learning, deployed at several labs internationally.
- Extending *datalab* to support additional domains, user workflows and modalities to enable research data management for automated, autonomous and otherwise data-driven labs.
- Pushing for interoperability in the chemistry and materials science software & data ecosystem via OPTIMADE and community efforts such as MADICES.

2024– **Scientific Software Consultant** and **Director**

datalab industries ltd.

- Customisation and deployment of *datalab* for industrial R&D and academic labs.
- Supporting the open source development of OPTIMADE and *datalab* via consultancy services.
- Key stakeholder in projects with partners spanning PSDI, CCDC, Faraday Institution and DIGIBAT.

2020–2025 **Research Assistant** (2020–2022) then **BEWARE Research Fellow** (2022–2025)

IMCN, Université catholique de Louvain and Matgenix SRL., with Prof Gian-Marco Rignanese

- High-throughput machine-learning accelerated workflows for materials discovery and design.
- Leading development of the OPTIMADE API specification and associated software (`optimade-python-tools`, `optimade-maker`).

2016–2020 **PhD student: Crystal structure prediction for next-generation energy storage**

University of Cambridge, with Dr Andrew Morris

- Computional materials discovery for conversion anodes for Li, Na and K-ion batteries.
- Author of two open-source Python packages: database approaches for high-throughput calculations and materials design with `matador` and crystal structure prediction with `ilustrado`.

OTHER EXPERIENCE

Visiting Researcher, Department of Chemistry, University of Cambridge (2021–2025); Postdoctoral Researcher, Cambridge Crystallographic Data Centre (2022); HPC Europa Visiting Researcher, Department of Applied Physics, Aalto University (2019); Scientific Software Developer Intern, Enthought Inc. (2019); Undergraduate Researcher, Department of Physics, University of Manchester (2014, 2015) & Department of Chemistry, University of Nottingham (2013)

SELECTED (AWARDS + HONOURS)

2025 Leverhulme Trust Early Career Research Fellowship with additional support from Isaac Newton Trust.

2022 BEWARE2 Fellowship from the Wallonia-Brussels Federation (approx. €300,000).

2021 PI for "Interoperable data management for fundamental battery research", BIG-MAP External Stakeholder Initiative (approx. €150,000).

SELECTED (TEACHING + SUPERVISION + SERVICE)

2026– Supervisor for 3 summer students (M. Bemana, S. Burra and J. Thomson)

2026– Data Champion at the University of Cambridge

2025– Co-organiser of the Research Software Engineering East of England (RS-Triple-E) group and events

2025– Co-organiser of the CECAM *OPTIMADE* workshop series, 2025 (Vilnius) and 2026 (Grenoble).

2025– Member of the steering committee for the Collaborative Computational Project for NMR Crystallography.

2022– Initiator and organiser of the CECAM MADICES workshop series in 2022 (virtual), 2024 (Berlin, Germany) and 2025 (Villigen, Switzerland).

2018– Reviewed for *Digital Discovery* (x8), *JOSS* (x7), *J. Phys.: Cond. Mat.* (x4), *Mach. Learn.: Sci. Technol.* (x3) (IOP Outstanding Reviewer 2024), *Sci Data* (x3), *npj. Comp. Mater.* (x2), *Sci. Rep.* (x1), *Sci. Technol. Adv. Mater.: Methods* (x1), *Physica Scripta* (x1), *Phys. Chem. Chem. Phys.* (x1)

2022–2024 Proposed and co-lead a MaRDA working group on metadata extractors for materials science.

2019-2021 Demonstrator: Part II Computational Physics, 3x Part IB Intro to Computing, Cavendish Laboratory

2016–2018 Supervisor: 2x Part IB Electromagnetism, Dynamics and Thermodynamics, Selwyn College

2012–2015 Tutor: GCSE Maths & Key Stage 2 Programming for The Tutor Trust, Manchester

SELECTED RECENT PRESENTATIONS

2026 4x Invited talks: AI4Materials, Henry Royce Institute, Manchester (March); AI4Materials, University of Oxford (April); AI and automation in chemistry, University of Cambridge (April); Advanced Battery Manufacturing Workshop, Cambridge (April); Swiss Battery Days, ETH Zurich (upcoming, August); RSC First: AI in Chemistry, Xiamen, China (upcoming, November)

Upcoming invited talk: Swiss Battery Days, ETH Zurich, Switzerland (August 2026)

2025 Invited talk: ML4Spectroscopy, Vrije Universiteit Brussel, Belgium.

4x Invited seminars: *Decentralized materials research data management, curation and dissemination for accelerated discovery*. MMDG, Uni. Birmingham; PSDI Polymer Data Workshop, Loughborough Uni.; Comp. Chem. Seminars, Uni. Warwick; AIChemy Seminars, Imperial College London

2024 Invited talk: *Decentralized materials research data management, curation and dissemination for accelerated discovery*, Symposium: Democratizing AI in Materials Science, MRS Fall, Boston, USA

Invited talk: *Federated, interoperable databases for accelerated materials discovery and design*, CECAM Flagship Workshop on MLIPs and Accessible Databases, Grenoble, France.

PUBLICATIONS

Underline indicates (joint) first authorship (reordered where appropriate). † indicates (co-)corresponding authorship. Full list with OA links available online (`https://ml-evs.science/papers`, ORCID, Google Scholar (950+ citations, h-index 14).

Preprints

23. Sutter-Fella, C. M., Ahmadi, M., *et al.* Robotic data management for trustworthy AI: A perspective from the perovskite semiconductor community, (2026). `10.26434/chemrxiv.15005693/v1`.

22. Roy, A., Shen, K., *et al.* From Knowledge to Action: Outcomes of the 2025 Large Language Model (LLM) Hackathon for Applications in Materials Science and Chemistry. *arXiv.org,* (2026). `10.48550/arXiv.2605.03205`.

21. **Evans, M. L.**†, Bocarsly, J. D., *et al. datalab*: federated data management infrastructure for materials chemistry and beyond. *ChemRxiv,* (2026). `10.26434/chemrxiv.15001945/v1`.

Peer-reviewed articles

20. **Evans, M. L.**†, Eimre, K., Rignanese, M. & Pizzi, G. `optimade-maker`: Automated generation of interoperable materials APIs from static datasets. *Digital Discovery,* (2026). `10.1039/D6DD00125D`.

19. Zimmermann, Y., Bazgir, A., *et al.* 32 Examples of LLM Applications in Materials Science and Chemistry: Towards Automation, Assistants, Agents, and Accelerated Scientific Discovery. *Mach. Learn.: Sci. Technol.,* (2025). `10.1088/2632-2153/ae011a`.

18. Trinquet, V., **Evans, M. L.** & Rignanese, G.-M. Accelerating the discovery of high-performance nonlinear optical materials using active learning and high-throughput screening. *J. Mat. Chem. C,* (2025). `10.1039/d5tc01335f`.

17. **Evans, M. L.**†, Rignanese, G.-M., Elbert, D. & Kraus, P. Metadata, automation, and registries for extractor interoperability in the chemical and materials sciences. *MRS Bulletin,* (2025). `10.1557/s43577-025-00925-8`.

16. Mroz, A. M., **Evans, M. L.**, *et al.* Cross-disciplinary perspectives on the potential for artificial intelligence across chemistry. *Chemical Society Reviews,* (2025). `10.1039/D5CS00146C`.

15. **Evans, M. L.**, Trinquet, V., *et al.* Optical materials discovery and design with federated databases and machine learning. *Faraday Discussions* **256,** 459–482, (2025). `10.1039/D4FD00092G`.

14. **Evans, M. L.**, Bergsma, J., Merkys, A., Andersen, C. W., *et al.* Developments and applications of the OPTIMADE API for materials data exchange and discovery. *Digital Discovery* **3,** (2024). `10.1039/D4DD00039K`.

13. Rosen, A. S., Gallant, M., *et al.* Jobflow: Computational Workflows Made Simple. *Journal of Open Source Software* **9,** 5995, (2024). `10.21105/joss.05995`. (2024).

12. Wang, Z., Gong, Y., **Evans, M. L.**, *et al.* Machine learning-accelerated discovery of A_2BC_2 ternary electrides with diverse anionic electron densities. *J. Amer. Chem. Soc.* **145,** 26412–26424, (2023). `10.1021/jacs.3c10538`.

11. Lertkiattrakul, M., **Evans, M. L.** & Cliffe, M. J. PASCal Python: A Principal Axis Strain Calculator. *Journal of Open Source Software* **8,** 5556, (2023). `10.21105/joss.05556`.

10. Jablonka, K. M., Ai, Q., *et al.* 14 Examples of How LLMs Can Transform Materials Science and Chemistry: A Reflection on a Large Language Model Hackathon. *Digital Discovery,* (2023). `10/gswbnx`.

9. Ells, A. W., **Evans, M. L.**, Groh, M., Morris, A. J. & Marbella, L. E. Phase transformations and phase segregation during potassiation of Sn_xP_y anodes. *Chemistry of Materials,* (2022). `10/h69d`.

8. **Evans, M. L.**, Andersen, C. W., *et al.* optimade-python-tools: a Python library for serving and consuming materials data via OPTIMADE APIs. *Journal of Open Source Software* **6,** 3458, (2021). `10/gn3w9f`.

7. **Evans, M. L.**, Andersen, C. W., *et al.* OPTIMADE, an API for exchanging materials data. *Scientific Data* **8,** 217, (2021). `10/gmnrxj`.

6. Breuck, P.-P. D., **Evans, M. L.** & Rignanese, G.-M. Robust model benchmarking and bias-imbalance in data-driven materials science: a case study on MODNet. *J. Phys.: Cond. Mat.* **33,** 404002, (2021). `10/gpw93d`.

5. **Evans, M. L.** & Morris, A. J. matador: a Python library for analysing, curating and performing high-throughput density-functional theory calculations. *Journal of Open Source Software* **5,** 2563, (2020). `10/gmf4mv`.

4. Harper, A. F., **Evans, M. L.** & Morris, A. J. Computational Investigation of Copper Phosphides as Conversion Anodes for Lithium-Ion Batteries. *Chemistry of Materials,* (2020). `10/gg5sx3`.

3. Harper, A. F., **Evans, M. L.**, *et al.* Ab initio Structure Prediction Methods for Battery Materials. *Johnson Matthey Technology Review* **64,** 103–118, (2020). `10/ggrmgf`.

2. Marbella, L. E., **Evans, M. L.**, *et al.* Sodiation and Desodiation via Helical Phosphorus Intermediates in High-Capacity Anodes for Sodium-Ion Batteries. *J. Amer. Chem. Soc.* **140,** 7994–8004, (2018). `10/gdq6h4`.

1. Zhu, T., **Evans, M. L.**, Brown, R. A., Walmsley, P. M. & Golov, A. I. Interactions between unidirectional quantized vortex rings. *Physical Review Fluids* **1,** 044502, (2016). `10/gf2529`.